

MGMT 298D: THE SCIENCE AND STRATEGY OF ARTIFICIAL INTELLIGENCE

Spring 2026 – 4 Units – In-Person

Course Syllabus

Teaching Team

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Time and Location

Course Day & Time: Mondays, 1:00-3:50pm
Course Location: G419
Course Site: BruinLearn

Description

This course demystifies artificial intelligence for a non-technical audience. We will go light on math and heavy on intuition. The goal is for you to build an appreciation for how underlying AI models *work* and how they unlock value from data in practice. We will explore various AI models using a mix of fully pre-written Python code and interactive web apps – no prior coding experienced is needed.

We will cover regression, tree-based models, clustering, reinforcement learning, neural networks / deep learning, image recognition, large language models (LLMs), and generative AI. We will consider use cases in demand forecasting, pricing, customer segmentation, personalized recommendations, autonomous vehicles, and more.

In the final two weeks, you will build an AI-powered micro-app and publish it on the web. Your goal will be to identify a specific business need addressed by AI and prototype a simple app, which will showcase your skills to potential employers and serve as a memento of the course.

Course Objectives

By the end of the course, students will be able to:

- 1. Frame business problems as prediction tasks for AI**
Understand how AI and machine learning models can be used to unlock value in a variety of use cases, including demand forecasting, pricing, customer segmentation, personalization, and image recognition.
- 2. Implement and apply AI methods to data**
Understand the intuitive foundations of AI models and gain hands-on experience with Python-based and no-code AI tools.
- 3. Build and pitch an AI-powered micro-app**
Collaborate in teams to conceptualize, prototype, and present an AI-powered micro-app that addresses a specific business need, integrating technical know-how with strategic thinking.

Learning Tools

We will primarily explore key AI concepts through user-friendly interactive web apps, allowing you to focus on conceptual understanding rather than coding. For demonstration purposes, we will sometimes use Google Colab's free notebook environment, which will allow us to implement AI models using (fully pre-written) Python code.

Optional Course Materials

[Book] *Deep Learning with Python*, Francois Chollet (2021)

[Book] *The Hundred-Page Machine Learning Book*, Andriy Burkov (2019)

[Book] *The AI Playbook: Mastering the Rare Art of Machine Learning Deployment*, Eric Siegel (2024)

[Video] *Deep Dive into LLMs like ChatGPT*, Andrej Karpathy (2025)

<https://www.youtube.com/watch?v=7xTGNNLPyMI>

[Video] *The spelled-out intro to neural networks*, Andrej Karpathy (2022)

<https://www.youtube.com/watch?v=VMj-3S1tku0>

[Website] *3Blue1Brown: Neural Networks* (2017).

<https://www.3blue1brown.com/topics/neural-networks>

[Book] *Neural Networks and Deep Learning*, Michael Nielsen (2019)

<http://neuralnetworksanddeeplearning.com/>

[Articles] Distill <https://distill.pub/> (Plain English explanations of AI research)

Course Outline

The course will be divided into three conceptual blocks as follows:

Weeks 1-3: **Machine Learning**

Weeks 4-7: **Deep Learning**

Weeks 8-10: **Product Sprint**

Week	Date	Title and Description	Deliverable
1	03/30/2026	Course Introduction; Foundations of Machine Learning <i>Use case: Retail Demand Forecasting</i> Course overview. Building intuition for model-fitting, using a case study on retail demand forecasting with H&M transaction data. We introduce core ideas for predictive modeling in high dimensions, including linear regression, LASSO regularization, train/test splits, under- and overfitting, and cross-validation. Students will build a demand forecasting model and explore how feature engineering improves accuracy.	A1 (Ungraded)
2	04/06/2026	Tree-Based Models: CART, Random Forests, and XGBoost <i>Use case: Algorithmic Pricing</i> This week focuses on advanced prediction models, using Carvana as a case study to illustrate the strategic differentiation enabled by algorithmic pricing. We cover classification and regression trees, random forests, and XGBoost. Students will build vehicle pricing models and compare performance across algorithms.	A2
3	04/13/2026	Reinforcement Learning <i>Use case: Dynamic Pricing; Recommendation Algorithms</i> This week focuses on dynamic decision-making, motivated by dynamic pricing and personalized recommendations. We introduce core ideas in reinforcement learning including state–action spaces, rewards, and the exploration–exploitation trade-off. Students will train a Q-learning agent and observe how decisions evolve with experience.	A3
4	04/20/2026	Neural Networks <i>Use case: Handwriting Recognition</i> Introduction of neural networks. We cover neural network architecture, including neurons, activation functions, hidden layers, stochastic gradient descent, and backpropagation. Brief comparison of NNs with linear and tree-based models. Students will build neural networks and experiment with different architectures and learning rates.	A4

Week	Date	Title and Description	Deliverable
5	04/27/2026	Computer Vision and Convolutional Neural Networks <i>Use case: Image Recognition and Autonomous Vehicles</i> Overview of computer vision applications in autonomous vehicles, medical imaging, and e-commerce. Students learn the basics of convolutional neural networks and how CNNs differ from dense networks. Students will train a CNN to classify different images and see a short in-class demo on object detection.	A5
6	05/04/2026	Transformers & Large Language Models <i>Use case: Sentiment Analysis on Yelp</i> Overview of the transformer architecture that has powered the AI revolution of the 2020s. We cover tokenization, embeddings, and self-attention, and explain how transformers enable modern LLMs. Students will train a small transformer model using thousands of Yelp reviews to predict ratings.	A6
7	05/11/2026	Generative AI <i>Use case: Investment Due Diligence</i> Focus on generative and agentic AI, using product-review analysis as a case study. Students learn how generative models produce text, based on concepts such as temperature and top-k sampling, and discuss how LLMs can operate as simple agents. Overview of fine-tuning and RAG. Students will use RAG to analyze the 10-K statements of three real robotics companies.	A7
8	05/18/2026	Agentic Patterns; Web App Development Tools <i>Use case: You Decide!</i> Overview of canonical agentic patterns and workflows. Turning machine-learning and AI ideas into lightweight applications. Students gain familiarity with LLM APIs (Gemini) and work in teams to build a small AI-powered web app.	N/A
9	05/22/2026	Prototyping and Publishing <i>Use case: You Decide!</i> Lab session to allow students to focus on development of their apps. Groups will meet with the instructor to review progress.	Product Proposal
10	06/01/2026	Product Pitches Product pitches of the top apps based on class voting.	Group Presentation

Evaluation and Grading

This course will be graded using the following weighted percentages for each of the components in the course. Feedback and grades are typically posted within one week of assignment due dates.

Assessment	% of Grade
Assignments [Individual]	60%
Micro-App Prototype [Groups]	35%
Peer Evaluation [Individual]	5%
Total	100%

Description of Assessments

Assignments (60%)

There are seven in-class assignments, designed to help students apply course concepts in practical settings using data and modeling tools. Assignments will be code-free and involve exploring a machine learning or AI method through an interactive web app. The best five out of seven will be counted (12% each).

Assignments will be submitted individually but collaboration is heavily encouraged. The course will include in-class time to complete the assignments, and submissions are due at 5pm the same day. Each assignment builds on technical and strategic skills relevant to AI model development, such as model building, evaluation, and interpretation.

Micro-App Prototype (35%)

The final project challenges students to prototype an AI-powered micro-app. Working in teams, students will identify a business use case, develop a working model or concept, and submit a short write-up. The project emphasizes the integration of business insight, technical design, and communication skills. The micro-app will be published on the web, with coding support from LLMs.

Peer Evaluation (5%)

A small component of each group's project grade will depend on evaluation by class. The top groups as identified by the class will pitch their products in Week 10.

Anderson and Course Policies

AI Usage Policy

This course will allow the use of AI (Large Language Models, including ChatGPT, Copilot, Gemini, and others) to enhance your learning, not replace it! The goal is to expand student capabilities and output in research and analysis, providing greater time to contemplate, analyze, and develop more sophisticated analyses, recommendations, and leadership imperatives. All students have access to Microsoft Copilot, the AI-powered chat mode of Microsoft using GPT 4 Turbo, a secure and protected tool once you login with your UCLA ID.

To facilitate the achievement of this goal, every assignment in the class will state whether using AI is appropriate. If you are confused about what is allowed, please ask the instructor or the course assistant for clarification. When using AI for individual or group assignments, you must acknowledge the tool's name and how it was used using the following style.

Netiquette

The written language has many advantages: more opportunity for reasoned thought, more ability to go in-depth, and more time to think through an issue before posting a comment. However, written communication also has certain disadvantages, such a lack of the face-to-face signaling that occurs through body language, intonation, pausing, facial expressions, and gestures. As a result, please be aware of the possibility of miscommunication and compose your comments in a positive, supportive, and constructive manner.

UCLA Policies

Code of Conduct

All participants in the course are bound by the **UCLA Student Conduct Code** (<https://deanofstudents.ucla.edu/individual-student-code>) and **UCLA Anderson Honor Code** (<https://www.anderson.ucla.edu/documents/areas/adm/web/AndersonHonorCode.pdf>).

Academic Integrity

UCLA is an institution of learning, research, and scholarship predicated on the existence of an environment of honesty and integrity. As members of the academic community, instructors, students, and administrative officials are all responsible for maintaining this environment. It is essential that all members of the academic community practice academic honesty and integrity and accept individual responsibility for their work. Academic misconduct is unacceptable and will not be tolerated in this course. Cheating, forgery, dishonest conduct, plagiarism, and collusion in academic misconduct erode the University's educational, research, and social roles.

Students who knowingly or intentionally conduct or help another student engage in acts that violate UCLA's expectations of academic integrity will be subject to disciplinary action and referred to the Dean of Students' Office.

Please familiarize yourself with **UCLA's Academic Integrity Policy**:

<https://www.deanofstudents.ucla.edu/Academic-Integrity>. Speak to your instructor if you have any questions about what is and is not allowed in this course.

Integrity in Research

Integrity in research includes not just the avoidance of wrongdoing, but also the rigor, carefulness, and accountability that are hallmarks of good scholarship. All persons engaged in research at the University are responsible for adhering to the highest standards of intellectual honesty and integrity in research.

Please familiarize yourself with the **University of California Policy on Integrity in Research**

(<https://www.ucop.edu/academic-personnel-programs/files/apm/apm-190-b.pdf>)

Accessible Education & Inclusive Education

Disability Services

UCLA is committed to providing a barrier-free environment for persons with documented disabilities. If you are already registered with the Center for Accessible Education (CAE), please request your Letter of Accommodation in the Student Portal. If you are seeking registration with the CAE, please submit your request for accommodation via the CAE website. Students with disabilities requiring academic accommodations should submit their request for accommodations as soon as possible, as it may take up to two weeks to review the request. For more information, please visit the CAE website (www.cae.ucla.edu), visit the CAE at A255 Murphy Hall, contact CAE by phone at (310) 825-1501, or by telecommunication device for the deaf at (310) 206-6083.

Equity, Diversity, and Inclusion

Please familiarize yourself with UCLA Anderson's commitment to maintaining an equitable, diverse, and inclusive community:

(<https://www.anderson.ucla.edu/about/equity-diversity-and-inclusion>)